PARDHASARADHI REDDY KARNATI

Software Engineer · Full-Stack Product Engineering · Applied AI

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SUMMARY

Software engineer who has shipped **five production systems as the sole engineer** — a multi-tenant SaaS platform, two customer web platforms, a cross-platform mobile app, a content and API backend, and an AI document-extraction engine. I own products end to end: system design, database schema, APIs, front end, deployment and live support. Equally comfortable in **React, Flutter, Python, Java and Node**, and in the applied-AI layer — **LLM integration, RAG, agentic workflows and real-time voice** — with a clear sense of where AI belongs in a system and where tested code should own the decision.

EXPERIENCE

Software Engineer · Dr. Swift Diagnostics

May 2025 – Present | Hyderabad

- Sole engineer for the company's entire product surface — five production systems across web, mobile, backend and AI, built from an empty repository through to live customer use.
- Designed multi-tenant architecture with **row-level tenant isolation, role-based access control and audited privileged actions** — the security model that lets one codebase serve many customer organisations safely.
- Integrated **LLMs into production workflows** — document extraction, RAG-grounded responses and agentic function calling — while keeping business-critical decisions in tested, deterministic code rather than model output.
- Established engineering practice for the team: automated **Playwright, Vitest and JUnit** suites, GitHub Actions CI/CD, shared design tokens across web and mobile, and written design-review checklists.

Project Intern · HCL Technologies

Aug 2023 – Dec 2023 | Chennai

- Shipped full-stack UI features and backend integrations across Agile sprints, coordinating with QA, design and backend teams for on-time releases.

SELECTED PROJECTS

Mr. Sales — Multi-Tenant SaaS for Pharma Field-Sales Teams

Three applications on one shared backend: a Flutter field app for five roles with GPS-verified visits and mock-location detection, a React tenant console with six-role RBAC, and a React operations console for subscriptions and invoicing. Designed the backend across 53 versioned migrations with row-level tenant isolation, plan-based entitlements and audit trails; offline writes are idempotent so retries never duplicate data.

Flutter, React, TypeScript, PostgreSQL / Supabase, RLS, Edge Functions, Vercel

AI Lab-Report Engine — Document Extraction & Visual Reporting Pipeline

Converts unstructured lab PDFs into branded, patient-readable visual reports and print-ready output. A Python service combines OCR with the Gemini API to produce schema-validated data; all clinical display logic lives in a unit-tested rule registry across nine panels, so no medical output depends on model behaviour.

Python, Gemini API, Tesseract OCR, PyMuPDF, Spring Boot, React, Playwright

drswift.in — Customer Web Platform & Backend-for-Frontend

Server-rendered TypeScript platform on Cloudflare Workers serving web, mobile and field-staff clients from one backend — booking and checkout, OTP and Google sign-in, and event-driven order sync with dead-letter handling and exactly-once HMAC-signed intake.

TypeScript, Cloudflare Workers, D1, R2, Queues, Firebase Auth · Also built cleocure.com (React, Vite, Cloudflare) end to end

SwiftCMS & Patient Mobile App — Content Backend and Cross-Platform Client

A Spring Boot content and operations backend serving a versioned public API with draft/live/retired publishing, audit logging and automatic CDN cache purging — consumed by a Flutter patient app (Android, iOS, web) with family booking, report viewing and biomarker trend charts.

Java, Spring Boot 3, JPA / Hibernate, Flutter, Riverpod, Thymeleaf

Real-Time Voice & Vision AI Agents — Production Conversational and Multimodal Systems

A production voice AI agent at sub-400ms latency on live telephony calls, with RAG-grounded responses and LangGraph agentic workflows triggering downstream actions from conversation. Extended into multimodal inspection combining computer vision, OCR and automated verification.

Python, Pipecat, WebSockets, LiveKit / WebRTC, LangChain, LangGraph, RAG, Docker / Kubernetes, AWS

TECHNICAL SKILLS

Languages TypeScript, JavaScript, Python, Java 17/21, Dart, SQL / PL-pgSQL, HTML, CSS
Frontend & Mobile React 18/19, Flutter, Vite, React Router, Riverpod, GoRouter, Tailwind CSS, Thymeleaf, design systems, WCAG accessibility
Backend & Architecture FastAPI, Flask, Spring Boot 3, Node.js / Express, Cloudflare Workers, REST API design, microservices, event-driven services, WebSockets, multi-tenant SaaS, RBAC & tenant isolation, HMAC signing, OAuth / OTP, audit logging, idempotent writes
AI / ML LLM APIs (OpenAI, Gemini, Claude), RAG, LangChain, LangGraph, agentic workflows, function calling, prompt engineering, OCR, computer vision, NLP, Transformers, CNNs, PyTorch, TensorFlow
Real-Time Voice Pipecat, LiveKit, WebRTC, SIP, Twilio, TTS / STT, speech-to-speech
Data PostgreSQL, Supabase (RLS, RPC, Auth, Edge Functions), MongoDB, Cloudflare D1, Firebase, JPA / Hibernate, schema migrations
Cloud, DevOps & Testing Docker, Kubernetes, AWS (EC2, S3), Cloudflare (Workers, R2, Queues, CDN), Nginx, Vercel, GitHub Actions CI/CD, Git, Linux, Playwright, Vitest, JUnit, flutter_test

EDUCATION & CERTIFICATIONS

B.E. Computer Science · Sathyabama Institute of Science and Technology, Chennai *2021 – 2025*
Certifications Google Generative AI, CDAC Cloud (AWS), Deep Learning, ML with MATLAB **Leadership** GDSC Contributor, Class Representative (4 yrs)